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Journal of Operations Management 16 (1998) 515–528

JOURNAL OF OPERATIONS MANAGEMENT

A population of learners: A new way to measure organizational learning

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Received 15 November 1995; accepted 24 August 1996

Abstract

In this paper we describe an approach that enables managers to systematically describe patterns of skill learning in large populations of front-line workers. The method fits individualized learning curves to the work performance histories of every member of a population. The resulting set of 'best-fit' parameter estimates represents a parsimonious summary of the many learning behaviors that are taking place. A case study demonstrated the descriptive power of the model using electronically collected data on 3874 episodes of individual learning in a large US manufacturing firm. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Learning curve; Empirical; Measurement; Visualization

1. Introduction

As manufacturers face increasingly intense competition, they must respond to a broader spectrum of market challenges. For example, managers and scholars are increasingly cognizant of the need for rapid learning throughout the organization. There is a growing consensus that organizations must become more effective learners if they are to contend with powerful external and internal forces. In particular, these environmental forces are thrusting many manufacturers toward greater 'agility', with attendant increases in product and service diversity and accelerating product and service innovation (Pine, 1993). Fig. 1 schematically shows how many companies have moved from making a few products with long commercial lifetimes to managing complex product families with multiple models and shrinking market lifetimes for individual designs and models (Uzumeri and Sanderson, 1995).

When companies fit the pattern in Fig. 1 (right hand side), the individual worker must constantly adapt to changing tasks. In industries like electronics assembly and apparel manufacturing, production lines that used to run for years are now reconfigured every few months. When this broad trend is combined with accelerating process innovation, greater cross-training and more aggressive restructuring of work through downsizing and outsourcing, it seems inevitable that future workers will have to master new tasks far more often. As depicted in Fig. 2, this trend promises to produce a work environment where

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PII: S0272-6963(97)00017-X

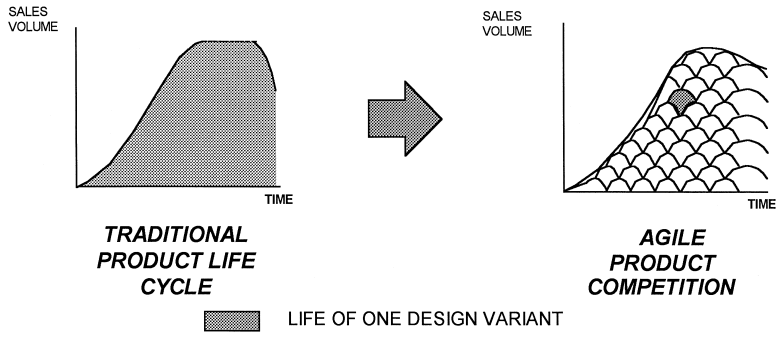

Fig. 1. Increased product variety and change.

workers spend much of their working lives on the steep part of one learning curve or another.

At the firm level, these frequent shifts mean that a growing fraction of the total workforce is on the steep part of the learning curve. Workers typically have minimal need for new information when they are performing familiar, routine tasks. When they are learning a new task, however, their need for information is prodigious. Worse, as workers operate increasingly costly and complex production processes, quality and productivity have become far more vulnerable to the confusion and indecision that comes from having too little information or experience. Thus, manufacturers are being challenged to deliver much larger quantities of timely, factual information so that workers can learn at their maximum rate and avoid the mistakes and confusion that rapid change too often creates. Accurate measurement is an essential step and this paper contributes to meeting this challenge by describing a new way to measure learning across large populations of workers.

The paper begins by reviewing the research that helped to define the proposed measurement strategy. The paper then examines a detailed case study where the proposed method is applied to data on manual assembly operations in a large US manufacturing organization. The results support both the concept of studying 'populations of learning curves' and the operational feasibility of measuring their learning behavior with real-world data.

2. Review of the literature

Scholars have studied learning behavior in organizations for more than a century. Important contributions have been made by behavioral psychologists studying the individual learning process, industrial engineers studying manufacturing costs, systems theorists studying organizational dynamics and artificial intelligence scholars trying to model human thought processes. Recently, behaviorists have added impor-

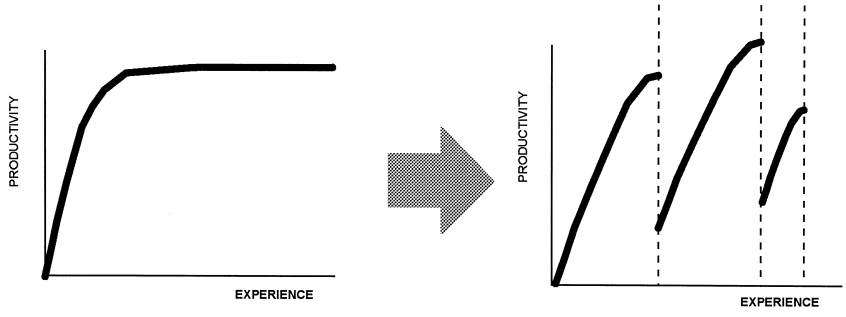

Fig. 2. Shift to truncated learning curves for individual workers.

tant insights into corporate culture and mechanisms for promoting and managing change. From this vast body of work, we have drawn primarily on three streams: Organizational experience curves, individual learning curves, and qualitative studies of learning organizations.

Learning curves for manufacturing efficiency and cost have been convincingly demonstrated many times over. Effective organizations use their experience to 'tune' operations to do the same work at a lower unit cost. Observers of aircraft manufacture in the 1930s noted that 'as the quantity of units manufactured doubles, the number of direct labor hours it takes to produce an individual unit decreases at a uniform rate.' (Yelle, 1979, pp. 302). Unfortunately, the observed cost reductions result from a complex mix of individual learning, technology changes, economies of scale, and changes in the management system (Dutton and Thomas, 1985, pp. 194). Thus, the internal structure and causation of organization-level learning curves are still poorly understood (Argote, 1993).

At the other extreme, the cognitive and physiological mechanisms by which individuals acquire knowledge have been carefully examined. Centered in behavioral psychology, artificial intelligence and industrial engineering, this research stream has generated sophisticated models of the processes by which humans acquire and retain knowledge (Anderson, 1982; Hancock, 1967; Mazur and Hastie, 1978). However, when the pressures to learn are driven by external change, many workers are affected simultaneously. There is currently no way to take individual learning process models and relate them to overall organizational learning. Therefore, some other intermediate approach is urgently required.

To fill the gap, various groups of scholars have used systems analysis, case studies, and observation to find ways for companies to increase the rate of organizational learning. This 'action research' agenda tries new approaches and observes their effects on organizational processes and outcomes (Argyris, 1989). These efforts have created a strong interest in helping employees to 'learn how to learn' (Argyris, 1982; Senge, 1992). A second stream of work has studied the economic impact of mid-level organizational learning. They show how diversity in learning behavior can exert a complex influence on overall productivity (Meredith and Camm, 1989; Adler, 1990; Kantor and Zangwill, 1991). Both types of intermediate level research have focused more on cognition and problem solving than on motor learning and skill acquisition. They reflect an implicit assumption that learning in professional and managerial ranks will have a greater impact on organizational and economic performance. In particular, scholars want to help organizations foster 'double loop learning,' where the decision-makers of an organization learn how to learn more effectively (Argyris, 1982).

Thus, there are still major gaps in our understanding of organizational learning, especially learning among front-line workers. These workers typically rely more heavily on skill acquisition and motor learning. While it is still possible to foster double loop learning (e.g., continuous improvement programs), the potential payoff is smaller than with complex decision making. Managers are more likely to be concerned with managing the *multiplicity* of learning behaviors that exist in a large workforce. In this situation, managers can do little to change the learning curves of specific individuals. However, by hiring selectively, assigning work appropriately and by terminating effectively, they can theoretically adjust the mix of learning curves for the overall workforce. To do this, however, one must first construct a tool that will allow scholars and managers to see and measure the multiplicity of individual behaviors.

3. The proposed approach

At the individual level, learning is affected by many factors, individual ability, individual variability, financial incentives, organizational norms and constraints, training, and the nature of the social environment. Learning and improvement also occurs in teams, work groups, departments, and other collections of people. In these groups, subtle interactions occur among the technologies, workers, and management systems that often lead to measurable performance improvements (Argote, 1993).

Until recently, quantitative analyses of these effects were prohibitively expensive. Now, however, computer technology allows organizations to record individual worker activities at shorter time intervals

at a dramatically lower cost. Much as the data from point-of-sale systems are transforming marketing research, the records from bar-code scanners and networked computers offer vast new mines of information on individual and group performance. To illustrate this, we have developed a two-step analysis approach:

Step 1: Fit a common mathematical function to time-series performance data for every individual in a given population.

Step 2: Use the resulting set of best-fit parameter estimates to describe the behavior of the entire group.

To apply this technique, one must have detailed, time-series data for a large number of individuals. [1] The performance of each individual must be recorded in a consistent form for all individuals. The technique also depends on finding a single mathematical function that can describe the observed patterns of learning. The functional form should be flexible enough to achieve a reasonable fit for as many individuals as possible. It should also be parsimonious (i.e., have as few parameters as possible) and be easy to estimate.

4. An illustrative case study

For the case study in this paper, we choose a model of individual learning advocated by Mazur and Hastie (1978). [2] They advocate a 3-parameter hyperbolic function that is flexible, easy to apply, and is known to reflect the way in which individuals learn both conceptual and motor skills:

$$y_{i,c} = k_i\left(\frac{c+p_i}{c+p_i+r_i}\right) + \varepsilon_{i,c} \qquad (1)$$

s.t. $c+p_i+r_i \neq 0$

In Eq. (1), $y_{i,c}$ is the work performance of individual i after c units of cumulative practice. The

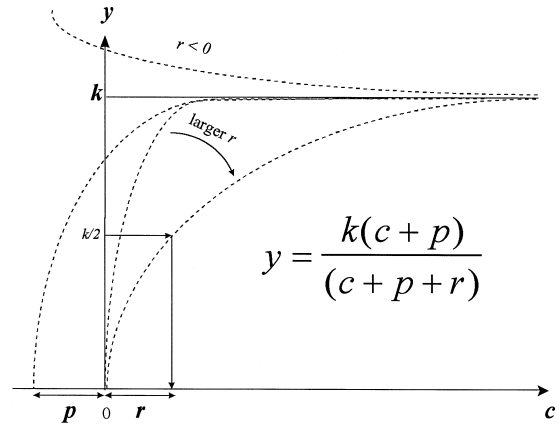

Fig. 3. The 3-parameter hyperbolic model of individual learning.

parameters k_i, r_i and p_i determine the shape of the improvement curves estimated from empirical performance data. As shown in Fig. 3, k_i estimates the asymptotic upper limit of $y_{i,c}$. In other words, k_i is an estimate of the maximum performance level that is projected when all improvement has taken place. The parameter p_i is an estimate of the accumulated prior experience of the individual in the same units as c. [3] The parameter r_i is a shape parameter that also has the same units as c. Mathematically, r_i is the cumulative work required to reach a productivity level equal to $k_i/2$. Thus, r_i is small when the initial improvement is rapid and is larger for improvement curves that start with a flatter slope. By adjusting k_i, r_i and p_i, Eq. (1) can adapt to a wide range of learning curve shapes.

Fig. 3 illustrates that the function can describe three distinctly different patterns of learning. First, if $r_i > 0$, the function rises to a limiting value of k_i. This is the case that one would expect to find most often in real-world work situations. The second situation occurs when r_i approaches 0 or p_i is very large. In this case, the function approaches a horizontal line, allowing it to describe situations where learning is not taking place. Finally, it can also describe the

[1] Glover (1966) made a reference to a similar idea, but did not pursue it empirically.

[2] In a detailed study that will be reported elsewhere, the authors compared Eq. (1) to a number of well-known learning curve models. For the data in the case study, the function in Eq. (1) exhibited less overall fit error and more consistent convergence than any of the alternatives.

[3] The value of c can be defined using any convenient units (e.g., work cycles, elapsed time) as long as the same units are used for every individual in the population.

situation (when $r_i < 0$ and $p_i > |r|$), where y declines for values of $c > -p_i$. Since $c \geq 0$ is true in all real-world situations, this third case can be used to describe situations where work performance actually declines over time.

The flexibility of the function lets it adapt to a broad spectrum of learning behaviors. For our purposes, this flexibility is more important than a strong causal basis, since the function must act as an unbiased estimator of the *shape* of the independent learning curves. This description must be as accurate as possible. The fact that the 3-parameter hyperbolic is easy to visualize and explain is a bonus.

4.1. The data

We applied the analysis technique to data from company A, a large US manufacturing company. The research data consists of work performance records taken at frequent intervals during 3874 *learning episodes* that occurred in company A over a one-year period from 1994 to early 1995. In each learning episode, a worker was asked to learn a specific assembly task. Each employee workstation had its own electronic terminal and bar-code reader to record his or her production output. The workers were paid on a piece-rate basis for work that passed inspection at subsequent stages of production. Defective units were not counted as output and were returned to the worker for repair. Thus, the output of the workers are somewhat quality weighted. At any time during their shift, employees could query their own terminal to see current earnings. This gave the workers nearly perfect knowledge of their current output rate and their current marginal pay rate. [4]

As each worker was asked to learn a new task, their training performance history was recorded in the database. [5] Each task was defined and classified by the industrial engineers of the company. In all, the data set contained records on 119 different tasks, 25 of which contained 30 or more learning episodes. The task definitions explicitly acknowledged that experience could be shared among the members of a class of similar tasks. The industrial engineers of company A had developed and refined a points system to explicitly estimate the overlaps in the various task types. We accepted those definitions in our analysis. [6]

At the start of the learning episode, performance data were collected approximately once per day. As the worker gained experience, the performance data were recorded as a weekly average. Altogether, the database contained nearly 68 000 individual performance records. Since the performance data determined the pay of the employees, it was carefully checked by employees and company management. Thus, we have a high degree of confidence in the reliability of the data.

4.2. Fitting the model

We fit the model in Eq. (1) to each learning episode using non-linear regression. [7] Since Eq. (1) increases smoothly over the relevant range, curve-fitting was straightforward and convergence was achieved in all 3874 episodes. [8] Fig. 4 illustrates fitted curves and underlying performance data for eight individuals, all learning the same task. In these graphs, y is expressed as a fraction of the standard production rate of company A for the specific task, which was set by the industrial engineers of the company. The horizontal axis, *cumulative work*, is the number of standard hours that the employee worked on the task in question. Since a standard hour is the time required to produce the standard amount of output, cumulative work is a measure of

[4] The combination of piece-rate pay and near-perfect knowledge of that pay eliminates the concern that the workers might have been uncertain about the performance objectives of the organization.

[5] The data was scaled to a common index to conceal the absolute values. In addition, the data and task characterizations were reviewed with the technical experts of company A to ensure that the descriptions were as fair and reasonable as possible.

[6] It is interesting to note that, whereas synergy has long been assumed in this factory environment, scholars have just recently recognized its economic impact in organization level decision-making (Meredith and Camm, 1989; Adler, 1990).

[7] SAS Procedure NLIN.

[8] Non-convergence occurred mainly when there were outliers or other badly-behaved performance data that caused the regression search algorithm to oscillate between two solutions. Even then, the differences tended to be quite small from an 'eyeball' perspective.

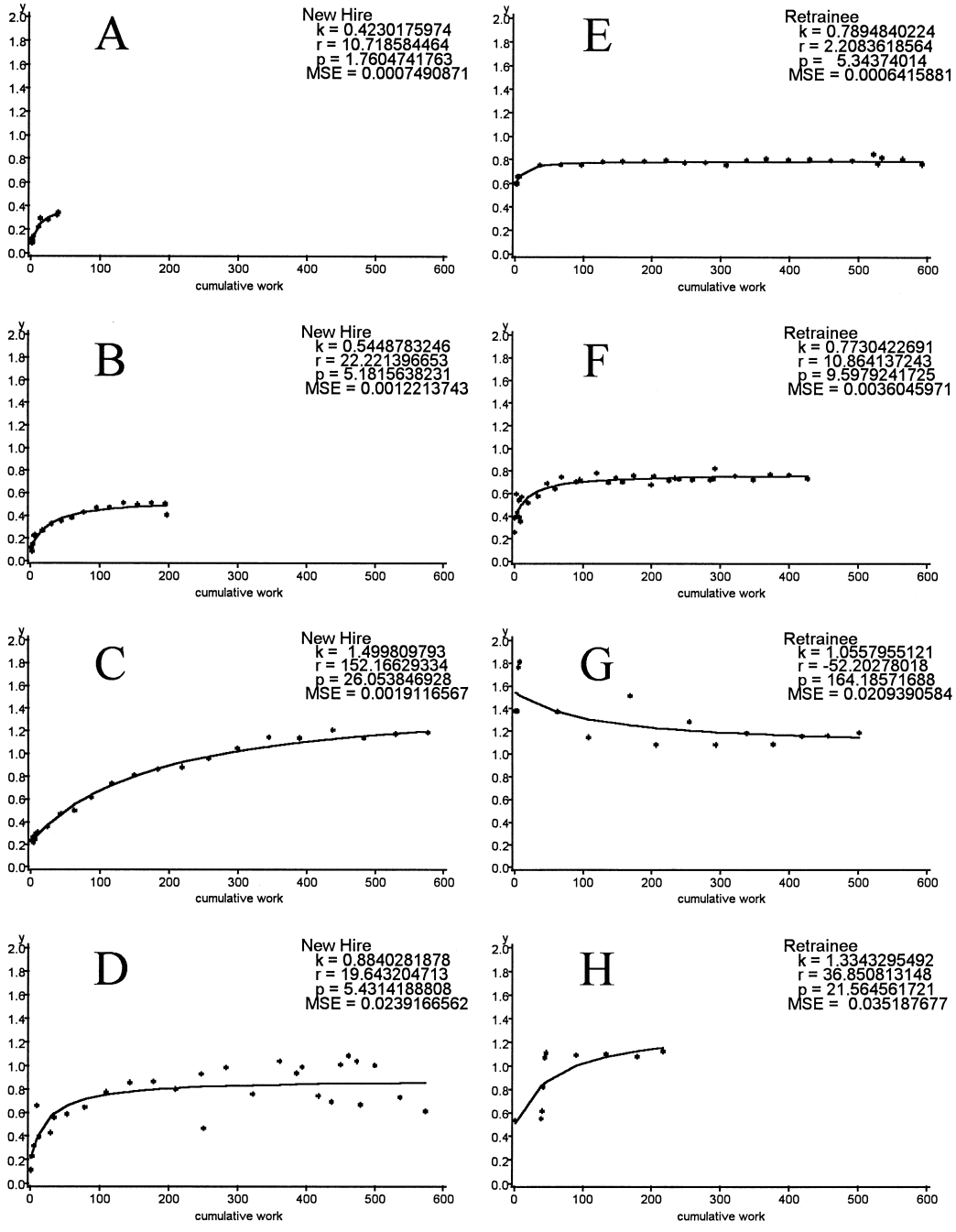

Fig. 4. Examples of fitted learning curves.

physical output and corresponds to the c in Eq. (1). From this, it follows that r and p are also measured in standard hours.

These examples were chosen because they span the range of data and fit quality for workers learning a specific task. Of the 3874 learning episodes, 347

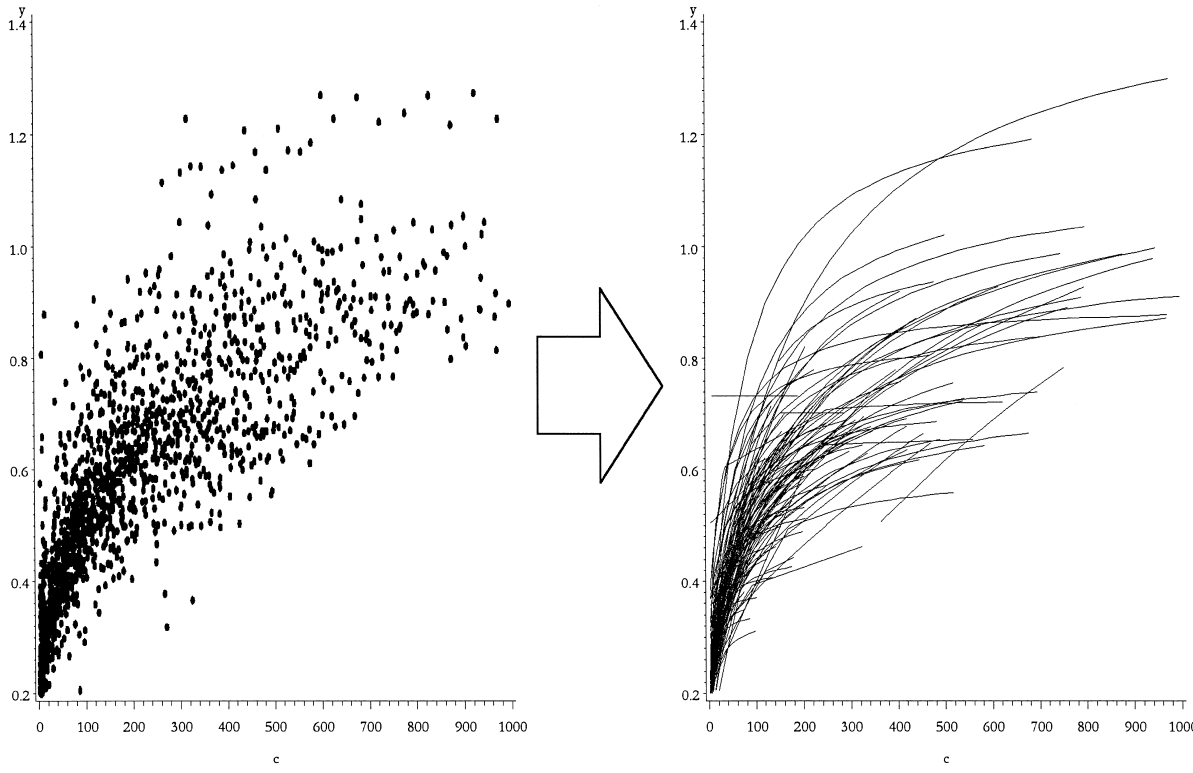

Fig. 5. Modeling a population of learners.

involved this one task. Of these, 110 episodes involved individuals who were newly hired and believed to have little or no prior experience in the industry. The other 237 episodes involved existing employees who were being retrained to perform this subtask. A review of the examples in Fig. 4 shows how variable real-world learning can be. For example, case A shows a newly hired worker who fails to measure up and either elects or is asked to leave the training process for this task. The task standard is $y = 1.0$, yet, this individual failed to reach $y = 0.4$ before vanishing from the data set. Since the data was comprehensive, departure is the only explanation for the truncated curve. [9] Case D shows a newly hired worker who makes rapid progress, but exhibits growing week-to-week variability as time goes by. The curve fitting seems quite good, but there is considerable unexplained variance. Case E shows a retrained employee who makes a very rapid adjustment to the task and subsequently performs with great precision. Case G shows a retrained employee whose performance actually falls over time. This illustrates the case where $r < 0$. Case H shows a situation where either the behavior is odd or the curve fit is questionable.

Limiting the choice of fitted curves to the function in Eq. (1) creates its own form of estimation error. One could always improve the fit for specific individuals by adjusting the functional form to achieve a closer fit. This is exacerbated by the fact that huge datasets are much harder to 'clean'. However, as long as there is no reason to suspect that the errors are biased, the set of parameter estimates k, p, and r can be seen as a parsimonious description of the overall learning behavior of the population. As shown in Fig. 5, this approach allows us to take a

[9] It is also possible that a learning curve may be right truncated because it extends beyond the study period. However, this did not occur in case A.

Table 1
Characteristics of fitted curves for task A ($n = 347$)

Measures	Percentile				
	10	25	50	75	90
Performance limit (k)	0.571	0.734	0.895	1.110	1.754
Shape parameter (r)	1.77	9.07	32.75	115.01	1060.5
Prior experience (p)	4.97	11.92	31.92	132.96	3562.9
Mean squared error (MSE)	0.0009	0.0016	0.0031	0.0053	0.0088

large quantity of raw data and convert it into a much smaller number of fitted equations.

Assessing goodness of fit for this approach posed several difficult problems. Normally, one would use residual analysis, regression diagnostics, and a general measure such as the R^2 statistic to guide the selection of variables to achieve the best fit to a given data set. In our application, however, the variables were set when the 3-parameter hyperbolic function in Eq. (1) was chosen. In this situation, variance can only be explained by the changing values of the three fitted parameter estimates. Thus, the meaning of conventional goodness-of-fit measures such as R^2 is less clear. The R^2 measure would be useful if we were selecting the best model to fit the data for a given individual. Here, however, the form of the model is fixed and we are applying it to data from many different subjects. Compare case D with case E in Fig. 4. Since *both* the total and the unexplained variance can vary between the two individuals, the ratio of the two has little meaning. Ultimately, our confidence in the goodness-of-fit derives mainly from a visual examination of a large number of graphs similar to those in Fig. 4. With few exceptions, all of the fitted curves met the 'eyeball' test. This was supported by studies of the *distribution* of fitted values and error terms for k, p, and r. A typical example of this, for a task designated 'A', is shown in Table 1.

It was reassuring that the fitted values of k seldom exceeded 1.5 since, historically, very few of the employees of company A ever exceeded this level. Further, the mean squared error for cases D and H in Fig. 4 lie at the upper end of the range listed in Table 1. Despite their relatively volatility, both fitted curves seem to be acceptable representations of the underlying data. More typical is the pattern of error in cases B, C, and F, which fall closer to the medians listed in Table 1. [10]

4.3. Looking for patterns

Once we had fit the learning curve model to the underlying data, we began to look for patterns in the best-fit parameters. Fig. 6 illustrates the fit for 110 episodes involving newly-hired workers attempting to learn one particular skill (task C). [11] Each performance history of an individual is summarized as a single point in the three-dimensional space defined by k, p, and r. A log transformation was applied to the parameter estimates for r and p to compress the scales and make the cloud more visible in the graphs and analyses that follow. [12]

Fig. 6 exhibits several interesting features. First, there is a small cluster of learning episodes that seem to reflect very poor performance. A review of the underlying data shows that these individuals resembled case A in Fig. 4, where the individual made an early departure from the training program. [13] A sec-

[10] MSE is the error sum of squares divided by $(n-3)$ degrees of freedom.

[11] By choosing newly hired workers, we avoided having to deal with situations like case G in…, where job performance drops over time. While the curve-fitting method was up to the challenge, it is difficult to construct a visualization of both learning and 'delearning' in the same diagram.

[12] The transformed values, $\log(r+1)$ and $\log(p+1)$, appeared to be distributed in a symmetrical, 'bell-shaped' distribution. The structure and meaning of these distributions is an area for ongoing investigation. Note that log refers to the natural logarithm throughout this paper.

[13] Most likely, they also left the company. This was inferred from the fact that they disappeared from the database before they had completed the training program and before the study 'window' had closed. The database was designed to retain their performance records as long as they remained with the company.

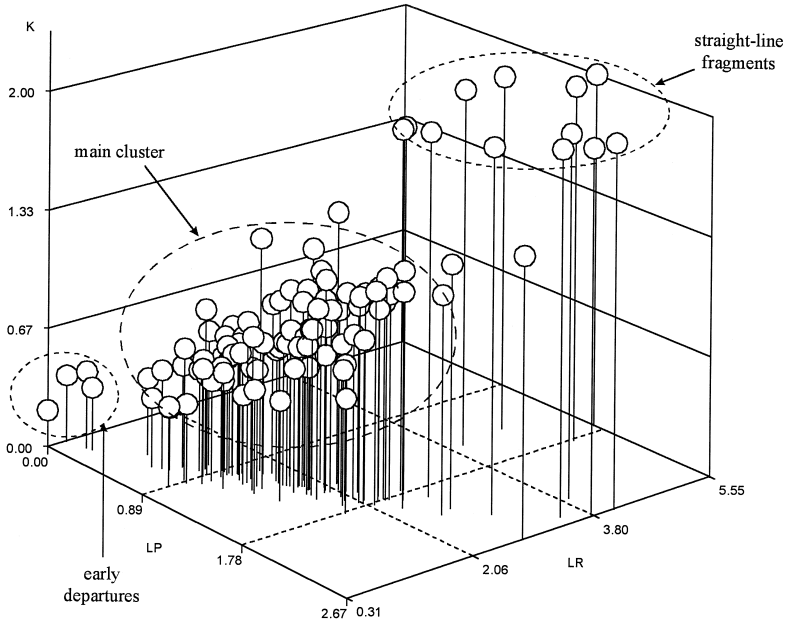

Fig. 6. Three-dimensional 'map' of organizational learning.

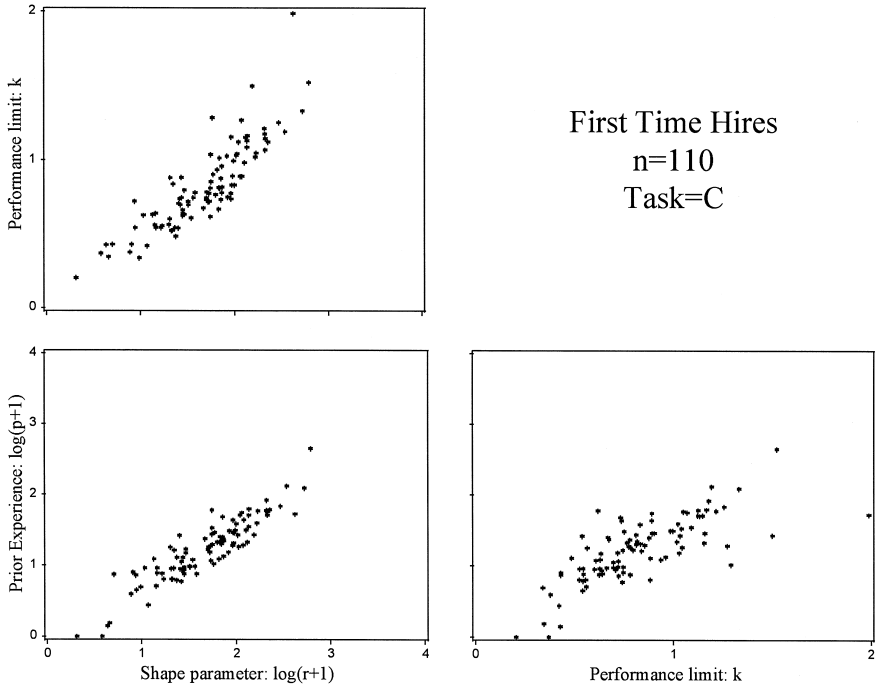

Fig. 7. Scatter matrix of fitted parameters estimates.

ond group of individuals exhibits large values for k, p and r. This pattern reflects individuals who enter training with substantial prior experience and, hence, exhibit little subsequent performance improvement (see case E in Fig. 4). In effect, these individuals may be miscast as novices.

For the remaining individuals, the center cluster reflects substantial variance in the best-fit parameter estimates. The values of k, for example, range from 0.5 of the standard output rate to 1.5 times the standard output rate. Some workers are projected to be nearly 3 times as productive as their colleagues at their ultimate steady state levels. Even allowing for the fact that these values are extrapolated, companies facing this pattern of variation would have a strong incentive to search for ways to distinguish among the different potentials early in the recruitment process.

Fig. 6 also suggests that a great deal might be learned by studying the *distribution* of improvement behavior across the population of workers. To visualize these patterns more easily, Fig. 7 depicts the parameter estimates in three orthogonal views.

The matrix illustrates that the three parameters are strongly correlated. This suggests that the shape of the improvement curves and the projected ultimate performance of individuals are systematically related and that certain learning behaviors are inherently unlikely. For example, it makes sense that workers with a high level of prior experience, p, (as indicated by a flat, non-zero improvement curve) would sel-

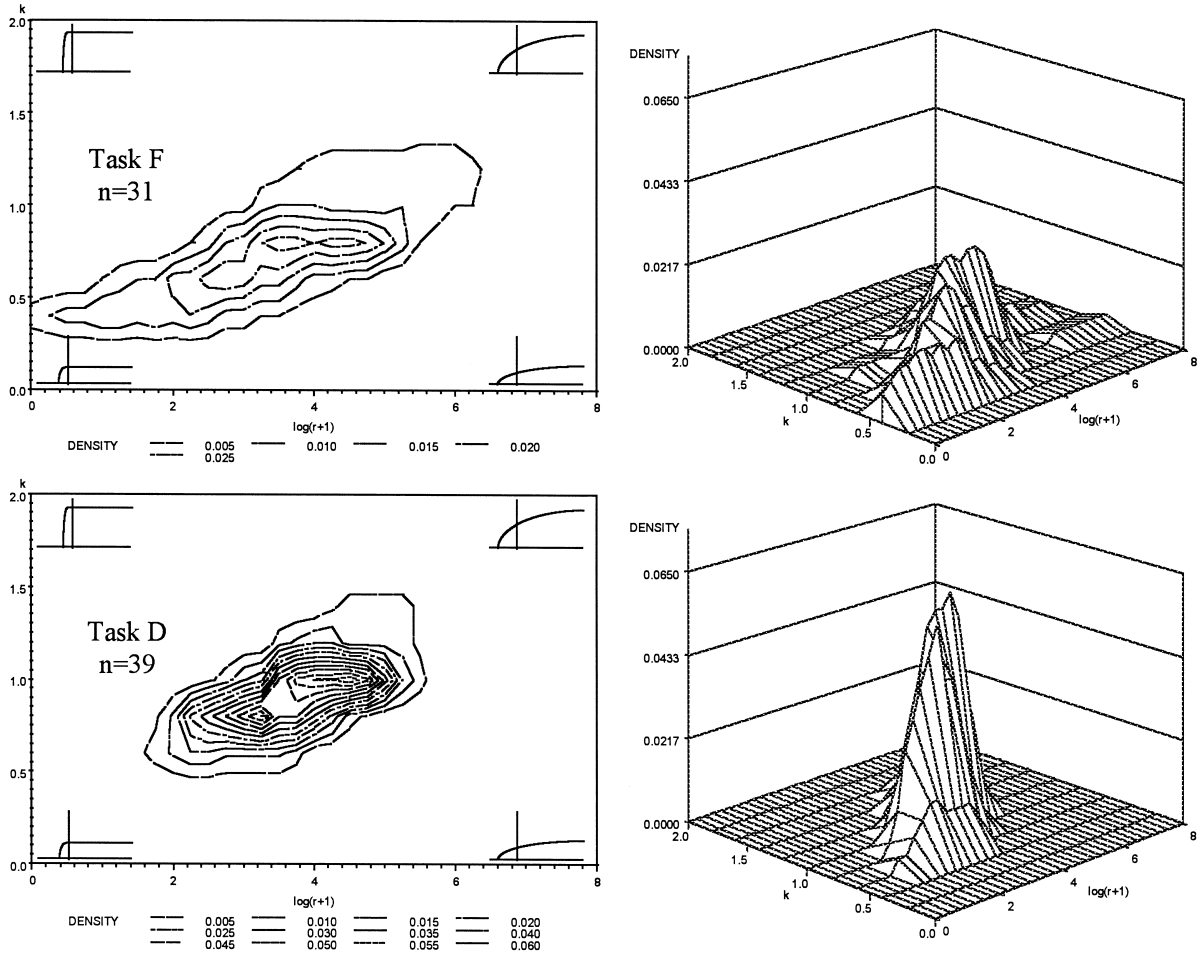

Fig. 8. Maps and surfaces for newly hired workers.

dom exhibit a low value of k. In order for those workers to have built the experience, they must have performed at or near the acceptable standard.

In general, Fig. 7 shows workers who tend to fall along a continuum. At one extreme, some workers learn quickly and then plateau at a relatively low level of performance. At the other extreme, some workers improve slowly and steadily, but climb to high ultimate performance levels. [14] While the reasons behind these patterns are beyond the scope of this paper, some of the potential explanations may relate to factors that are very site-specific. For example, the rapid initial rise among lower-performing workers (point B) may indicate that some workers already have skills but dislike the work. Alternatively, it may indicate that some workers find the task easy at first and then become frustrated.

Regardless of the reasons, a company can use these distributions to create a powerful empirical description of its workforce. In the examples that follow, we focus on maps that juxtapose k and r. This combination shows how the time required to acquire a skill (r) correlates with the likely ultimate level of skill attainment. Fig. 8 shows two versions of this type of map, applied to learning on two different tasks.

In Fig. 8, the contour maps on the left show the density of k vs. $\log(r+1)$ combinations. The densities resemble a probability density function for the observed combinations. [15] The relief in the contour plots is visualized in the corresponding surface plot. In these examples, as well as plots for other tasks in the full data set, different tasks can have very different maps. For example, task D exhibits a different mode and a lower variance than task F. Workers seem to learn task D in a more consistent manner than task F. However, since the maps do not indicate *why* these patterns exist, explanations of learning behavior must come from other sources. However, maps like the one in Fig. 8 may reveal patterns that can tell managers what areas for further study are likely to be most fruitful.

5. A statistical distribution of learning behavior

In a broad sense, the 'learning maps' can help any organization to build a quantitative description of learning within its workforce. However, the applications extend beyond managerial curiosity. In our efforts to apply the model to the data in company A, we see particular promise in the possibility that we can describe groups of learners in quantitative statistical terms.

Organizational learning is a group characteristic and most managerial interventions (e.g., education, training, and technology) are designed to change group performance. Managers tend to be more concerned about the *mix* of learning curve shapes in the workforce than in changing the curves of specific individuals. To deal with individual curves, one would need to abandon the power of statistics and understand the many subtle forces acting on that specific individual. In other words, it is more important to change the group learning map than it is to change the behavior of specific workers.

In focusing on the group level, the distribution of learning curve shapes may provide clues to systemic influences that would otherwise be very hard to see. The regularity in our examples of the learning map suggests that, while the parameter estimates are highly variable across individuals, they may be reasonably predictable in the aggregate. This result means that managers and scholars can use common statistical modelling and analysis tools to describe and apply the distributions involved.

To illustrate this, Fig. 9 shows four learning maps for learning on two different tasks by two different groups of employees. One group contains newly-hired workers while the other represents current employees who are being retrained or cross-trained for the tasks in question. The maps raise several questions. Why, for example, is there less variance in the learning curves for newly hired trainees than for employees who change jobs? Why is the most com-

[14] Similar patterns were found in 12 other task categories that were examined in detail.

[15] The frequencies were measured from data points like those shown in Fig. 5. The densities are calculated using the non-parametric density estimation techniques in SAS PROC DISCRIM. For each point, the neighbors within a fixed-radius sphere are counted and divided by the volume of the sphere. The resulting density surface is scaled to make the area under the surface equal to 1.

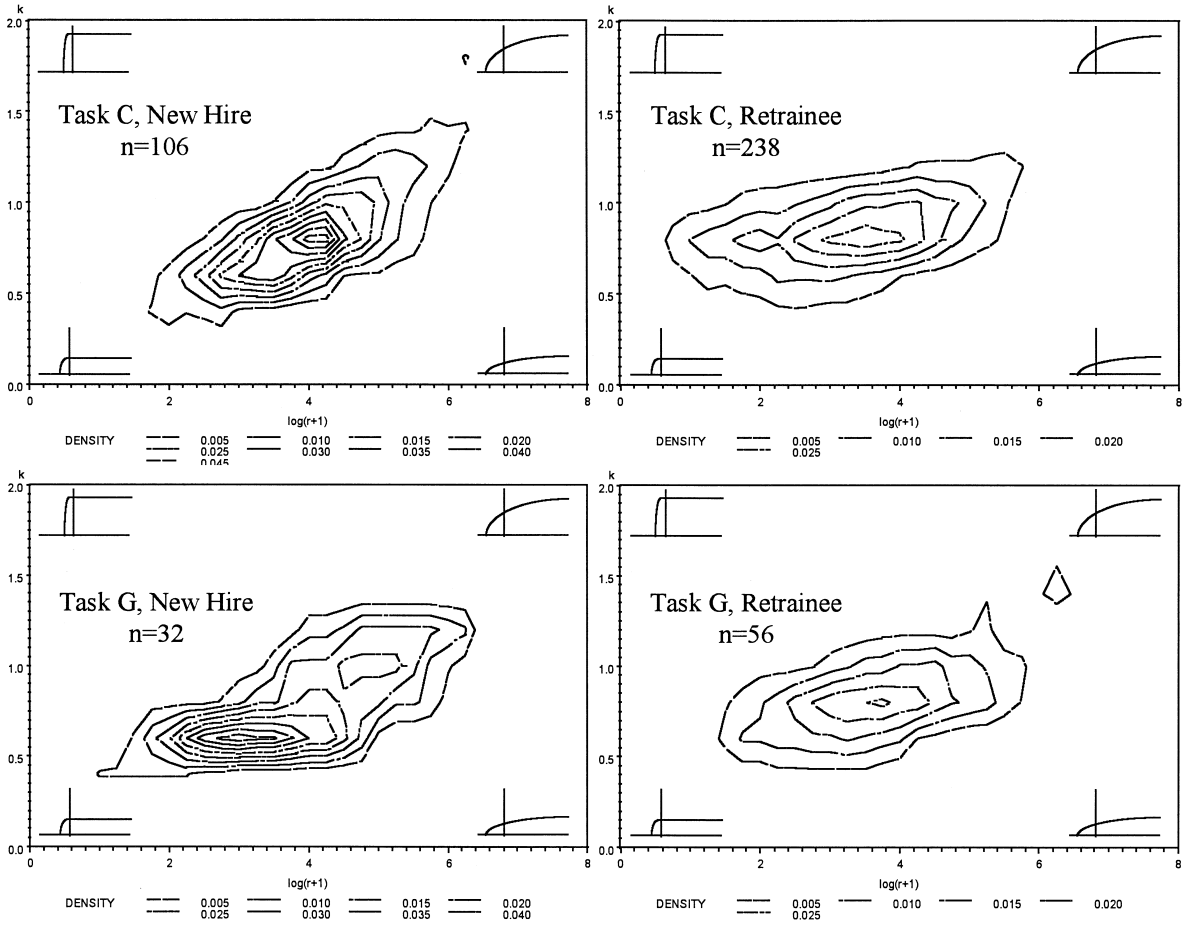

Fig. 9. Typical improvement maps.

mon value of k lower for the newly hired workers on task G than for the current employees?

These questions would be difficult to visualize with traditional aggregate modelling approaches. By viewing an organization as a *population* of learning curves, one can implicitly 'filter out' many of the variations in performance that result from minor day-to-day effects. With these eliminated, what remains are the patterns of variation in the long-term learning behavior of different individuals.

Shifts in the distribution of the parameter estimates may help managers to identify when external forces are systematically impacting the learning performance of the organization. Comparisons of successive maps may indicate if learning is accelerating or eroding and whether the ultimate prospects for performance are increasing or decreasing. Comparisons of maps across operating units may uncover differences in learning behavior in different locations, at different tasks, at different times, or among workers with different types and levels of training.

We anticipate that these empirical distributions will be particularly useful in situations where rapid organizational change is taking place. For example, a company opening a plant in an unfamiliar region could estimate maps based on early performance data at the new plant. By comparing the preliminary map to those of mature plants, the company could identify systematic differences in workforce adjustment before they are visible through other means.

The maps may also help management to better understand the diversity of their workforce. This

could create new opportunities to take patterns of worker variability into account in making key decisions. Consider a company that is introducing a new product line. Should the company make the product in a new plant with new workers? Should it make the product in an old plant with experienced workers and shift other products to the new plant or should it simply expand its existing plant and use a mix of experienced and inexperienced workers?

For task C in Fig. 9, the new workers seem to have a higher maximum productivity than retrained workers, but they pick up the new skill more slowly. [16] If startup productivity is a big issue, or if the new product is likely to be redesigned fairly often, it may make sense to maximize the use of the experienced workers. If the product is likely to have a long product life, it might be better to bet on the long term potential of the new work force. The company could even input the learning maps in Fig. 9 as prior probability distributions in a simulation of the production process for each of the three alternative settings.

Finally, it is important to reiterate that the fitted curves are a fundamentally *descriptive* model of learning behavior that does not need to assume a specific causal mechanism. When used in this manner, descriptive curves can be fit to any variable. Instead of production rates, maps could be drawn for reductions in defect rates, for changes in cycle times, or for improvements in worker safety. While not every company has production records as detailed as Company A, most collect at least some variables that could provide a plausible basis for mapping organizational learning.

6. Summary

In this paper, we have outlined a way to quantitatively describe the behavior of learning and improvement processes across populations of front-line workers. This approach reflects our view that emerging information technology will soon be sufficiently widespread that many companies will have the raw data to estimate reliable improvement curves for individual employees on well-defined tasks. Companies can use the resulting fitted parameter estimates as a 'shorthand' to describe learning behavior of large groups of individual workers.

To demonstrate the feasibility and application of this approach, we conducted a case study with a large quantity of data from a large US manufacturing company. This empirical study revealed intriguing patterns of regularity and variance. Potential applications of this method include: The integration of learning into product costing, the use of learning maps to gauge training effectiveness, the comparison of learning maps in making plant location and investment decisions, the use of learning maps to improve operational flows, and the use of learning maps to identify potential and areas of organizational dysfunction and confusion.

We hope that follow-on studies will lead to improvements in both the method and in our understanding of systematic patterns in the learning of large groups of people. In particular, we hope that a better characterization of the *distribution* of learning behavior will allow managers and scholars to build new decision-making tools that can integrate the real world multiplicity of learning behaviors into broader organizational and economic decisions.

[16] As with many of the examples, the reasons behind these patterns can only be determined through site investigations, which the confidentiality requirements of company A preclude us from discussing. Hypothetically, however, this pattern might result if newly hired workers were more focused and energetic than the existing, more experienced workers.

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